

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM: 4**

**TERM: 3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: LOCOMOTORY FEATURES OF A FINNED FISH.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify the locomotory features of a finned fish.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Begin with a warm greeting to engage students and create a positive learning atmosphere.
- Introduce the topic by explaining the importance of locomotion in fish and its relevance to their survival.
- Mention that a freshly killed tilapia will be used for observation today to better understand its locomotory features.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- A freshly killed tilapia
- Textbook

**Learning Activities:**

- Observing external features of tilapia
- Drawing and labeling the tilapia
- Participating in a group discussion

**Lesson Content (30 minutes):**

- Begin by having students observe the external features of the freshly killed tilapia. Point out key features such as fins, tail, and body shape.

*Discuss how these features contribute to the fish's ability to swim effectively:*

- Fins (dorsal, pectoral, pelvic, anal, and caudal fins) help with steering and stability.
- The streamlined body shape minimizes water resistance, allowing for swift movement.
- The tail (caudal fin) acts as a primary propelling force when moving through water.
- Encourage students to draw the fish and label the different locomotory structures, emphasizing their functions.

- Facilitate a discussion on the differences in locomotion among different fish species and how various environments may affect these adaptations.
- Conclude this section by reiterating vital points, and asking questions to assess understanding.

**Resources:**

- Textbook
- Digital resources (if available) on finned fish locomotion
- Freshly killed tilapia for observation

**Learning Activities:**

- Observation of fish features
- Drawing and labeling diagrams
- Group discussion on locomotion adaptations

**Conclusion (5 minutes):**

- Summarize the key points discussed in the lesson, highlighting the importance of fins and body shape in fish locomotion.
- Emphasize how these features are adaptations for life in an aquatic environment.
- Prepare students for the next lesson by hinting at an upcoming exploration of locomotion in other aquatic animals and comparing it with that of finned fish.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Summary discussion
- Question and answer session to clarify any doubts

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: LOCOMOTORY FEATURES OF A FINNED FISH.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify the locomotory features of a finned fish.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students warmly and create an engaging atmosphere for learning.
- Introduce the topic by explaining the importance of locomotion in fish and how their unique adaptations help them move efficiently in water.
- Engage students with a quick question to provoke thought (e.g., “How do you think fish swim differently from land animals?”).

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Quick discussion on students' prior knowledge on locomotion.
- Presentation of the resource: freshly killed tilapia.

**Lesson Content (30 minutes):**

- Start with a hands-on observation of the tilapia, leading students to note the external features that contribute to locomotion such as fins (pectoral fins, pelvic fins, dorsal fins, anal fin, caudal fin) and body shape.
- Encourage students to gather around and observe the tilapia closely, demonstrating how to identify each fin and discussing its specific role in swimming.
- Have students draw the tilapia, labeling its fins and other relevant structures. Discuss how these adaptations help the fish navigate, accelerate, and change direction in water.
- Facilitate a group discussion where students can share their observations and thoughts about how the features relate to locomotion. Summarize their findings, connecting the biological structure to function.

**Resources:**

- Freshly killed tilapia
- Chalk board/white board
- Textbook
- Drawing materials (paper, colored pencils)

**Learning Activities:**

- Hands-on observation of tilapia
- Drawing and labeling of the fish structures
- Group discussion of observations and conclusions

**Conclusion (5 minutes):**

- Summarize the key points by reiterating the essential locomotory features of a finned fish and their importance in aquatic environments.
- Encourage students to think about how the adaptations discussed today might differ across various species of fish or other aquatic animals.
- Announce the topic for the next lesson so students can prepare in advance.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Summary of key points
- Preparation for the next lesson

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: LOCOMOTION IN A FINNED FISH.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

### **Objectives:**

By the end of the lesson, the learner should be able to:

1. Explain how locomotion occurs in a finned fish.
2. Explain how a fish is adapted to locomotion in its habitat.

### **LESSON PRESENTATION**

#### **Introduction (5 minutes):**

- Greet the students warmly and introduce the lesson topic, "Locomotion in a Finned Fish".
- Engage students by asking them if they have seen fish swimming and what they think allows them to move in water.

#### **Resources:**

- Chalkboard/whiteboard
- A piece of chalk/marker pen
- Freshly killed tilapia
- Textbook (KLB BK IV, pp. 117-118)

#### **Learning Activities:**

- Discussion on external features of fish
- Introduction to the concept of locomotion

#### **Lesson Content (30 minutes):**

##### *1. Review external features of a tilapia (10 minutes):*

- Use the freshly killed tilapia to point out key external features such as fins (pectoral, pelvic, dorsal, caudal), scales, and body shape.
- Discuss how these features contribute to swimming efficiency, maneuverability, and stability in water.

##### *2. Detailed Discussion on Locomotion (20 minutes):*

- Explain the mechanics of fish locomotion, focusing on how fin movements and body shape facilitate movement.

- Discuss the role of muscles and the skeletal system in locomotion and how fish alternate fin propulsion and body undulation.
- Highlight adaptations such as streamlined bodies for reduced drag, and the role of the lateral line system in sensing water currents and obstacles.
- Encourage group discussions where students can share what they learned about how a fish is adapted to its environment.

**Resources:**

- Freshly killed tilapia
- KLB BK IV, pp. 117-118
- Multimedia presentation (if available)

**Learning Activities:**

- Group discussion on external features
- Hands-on observation of the tilapia
- Interactive Q&A on locomotion mechanics

**Conclusion (5 minutes):**

- Summarize key points: the fins' role in movement, the importance of body shape for swimming, and special adaptations of fish for efficient locomotion.
- Prepare students for the next lesson by mentioning that they will explore other forms of locomotion in different animals.
- Encourage questions for clarification.

**Resources:**

- Chalkboard/whiteboard
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Recap of the lesson's main points
- Q&A session

**Teacher Self-Evaluation:**

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**FORM:4**  
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**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: LOCOMOTION IN A FINNED FISH.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

1. Explain how locomotion occurs in a finned fish.
2. Explain how a fish is adapted to locomotion in its habitat.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson with a warm greeting to create a positive learning environment. Introduce the topic of locomotion in finned fish by discussing its importance in their survival and adaptation to aquatic life. Pose a question to spark interest, such as: "How do fish move so gracefully in water?"

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Class discussion
- Questioning

**Lesson Content (30 minutes):**

*1. Review External Features of a Tilapia (10 minutes)*

- a. Utilize the freshly killed tilapia to examine its external features. Discuss the significance of features like fins, streamlined body shape, and the caudal fin (tail) for locomotion.
- b. Students will observe and identify how these features aid in movement.

*2. Detailed Discussion (20 minutes)*

- a. Explain how locomotion occurs in a finned fish, focusing on the function of muscles and fins during swimming. Describe the undulating movements that create propulsion.
- b. Discuss the adaptations of tilapia that facilitate movement in water:
  - Body shape (streamlined) reduces resistance.
  - Fins (pectoral, pelvic, dorsal, anal, and caudal) guide movement and balance.

- Musculoskeletal system helps in powerful movements.

c. Group discussion: Divide students into small groups to discuss how these adaptations help tilapia thrive in their aquatic habitat, and ask them to share their findings with the class.

**Resources:**

- KLB BK IV, PP 117-8
- Freshly killed tilapia
- Chalk board/white board
- Textbook

**Learning Activities:**

- Hands-on examination of a tilapia
- Group discussions
- Class presentations

**Conclusion (5 minutes):**

Summarize the main points covered in the lesson, emphasizing the relationship between the anatomical features of the tilapia and its ability to swim effectively in its habitat. Remind students of the adaptations that enhance fish locomotion. Prepare students for the next lesson by hinting at the next topic related to animal movement.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Recap discussion
- Question-and-answer session

**Teacher Self-Evaluation:**



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**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: LOCOMOTION IN A FINNED FISH.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

### **Objectives:**

By the end of the lesson, the learner should be able to:

1. Explain how locomotion occurs in a finned fish.
2. Explain how a fish is adapted to locomotion in its habitat.

### **LESSON PRESENTATION**

#### **Introduction (5 minutes):**

- Greet the students warmly and engage them with a brief discussion about different types of fish they know.
- Introduce the topic of locomotion in finned fish by highlighting its importance in helping fish navigate their aquatic environment efficiently and effectively.

#### **Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook (KLB BK IV. PP 117-8)
- A freshly killed tilapia

#### **Learning Activities:**

- Brief discussion on types of fish
- Overview of the importance of locomotion

#### **Lesson Content (30 minutes):**

- Begin with a review of the external features of the tilapia. Use the freshly killed specimen to point out and explain the role of fins, body shape, and tail in locomotion.
- Discuss how fish use their body and fins to generate movement in water, making connections to concepts such as lateral movement and the role of muscles.
- Explore how the specialized fins (pectoral, pelvic, dorsal, anal, and caudal fins) contribute to balance, steering, and propulsion.
- Highlight adaptations such as streamlined body shape, flexibility, and the arrangement of muscles which facilitate effective locomotion in their natural habitat (e.g., rivers, lakes, oceans).

- Conduct a hands-on activity where students analyze the tilapia's body features and relate each feature to its function in locomotion.
- Engage students in a question-and-answer session to reinforce their understanding and promote critical thinking regarding the adaptations of fish to their environments.

**Resources:**

- KLB BK IV. PP 117-8
- Freshly killed tilapia

**Learning Activities:**

- Review of external features
- Detailed discussion and explanation of locomotion
- Hands-on analysis of tilapia
- Q&A session

**Conclusion (5 minutes):**

- Summarize the key points covered in the lesson, reiterating the mechanisms of locomotion in fish and the adaptations that facilitate this movement.
- Emphasize the significance of these adaptations for survival in aquatic environments.
- Prepare students for the next lesson by giving a preview of the content they will cover.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook (KLB BK IV. PP 117-8)

**Learning Activities:**

- Summary of key points
- Preview of the next lesson

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: TAIL POWER OF A FISH**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to calculate the tail power of a fish.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson with a warm greeting to engage the students. Introduce the topic of the tail power of fish, explaining its significance in fish locomotion and survival in aquatic environments. Highlight that the tail plays a crucial role in movement and navigation in water.

**Resources:**

- Chalkboard/whiteboard
- A piece of chalk/marker pen
- KLB BK IV, PP 118-9

**Learning Activities:**

- Brief discussion on the role of the tail in fish movement.
- Introduction to the concept of tail power.

**Lesson Content (30 minutes):**

*1. Discussion on Tail Structure and Function (10 minutes):*

- Explain the anatomy of a fish tail, including its parts (caudal fin, musculature).
- Discuss how the tail structure relates to its function in propulsion and maneuverability in water.

*2. Measuring and Calculating Tail Power (15 minutes):*

- Instruct students to measure the length of the entire tilapia fish and then just the tail.
- Introduce the formula for calculating the tail power of a fish, incorporating parameters such as body length and tail length.
- Guide students through hands-on calculations using the measurements they obtained. Provide them with the necessary formula for calculation.

*3. Discussion on the Significance of Tail Power (5 minutes):*

- Engage students in a discussion about how tail power affects fish locomotion, predator evasion, and environmental adaptation.
- Encourage students to think about how different species of fish might have evolved different tail shapes and sizes based on their habitats and lifestyle.

**Resources:**

- Freshly killed tilapia for measurement
- Rulers or measuring tapes
- KLB BK IV, PP 118-9

**Learning Activities:**

- Group activity measuring fish lengths.
- Calculation of tail power using the formula (example provided in text).
- Class discussion on real-life implications of tail power in different fish species.

**Conclusion (5 minutes):**

Summarize the key points covered in the lesson, reiterating the importance of the tail in a fish's movement and survival. Mention the calculations they performed for tail power and its implications in the study of fish biology. Prepare students for the next lesson, which will expand on how different types of locomotion are adapted in various aquatic species.

**Resources:**

- Chalkboard/whiteboard
- A piece of chalk/marker pen
- KLB BK IV, PP 118-9

**Learning Activities:**

- Summary of key points discussed.
- Q&A session to clarify any doubts.

**Teacher Self-Evaluation:**

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**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: TAIL POWER OF A FISH.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

1. Calculate the tail power of a fish.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students warmly and explain the significance of studying the tail power of fish within the context of locomotion in aquatic environments.
- Introduce the concept of "tail power" as a measurement of the effectiveness of the tail in propelling fish through water.
- Inform students that they will be measuring and calculating the tail power of a freshly killed tilapia fish as part of their learning.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook (KLB BK IV, PP 118-9)
- A freshly killed tilapia

**Learning Activities:**

- Introduction and class discussion
- Individual measurements
- Calculations

**Lesson Content (30 minutes):**

- Begin by discussing the anatomy of the tilapia, focusing on the structure and function of the tail and its role in movement.
- Divide students into pairs and instruct them to measure the length of the tilapia fish and the length of the tail using a ruler.
- Guide the students on how to calculate the tail power using the formula: Tail Power = (Tail length / Total body length)  $\times$  100.

- Encourage students to compare their results with their peers and discuss any differences they observe.
- Facilitate a class discussion on the significance of tail power in locomotion. Ask guiding questions such as:
  - Why is tail power important for a fish's survival?
  - How does tail power affect swimming speed and maneuverability?
- Summarize the calculations and observations made during group activities.

**Resources:**

- KLB BK IV, PP 118-9
- Ruler
- Calculation materials (paper, calculators)
- Freshly killed tilapia

**Learning Activities:**

- Group measurements and calculations
- Class discussion

**Conclusion (5 minutes):**

- Recap the critical concepts discussed in the lesson, emphasizing the calculation of tail power and its relevance in aquatic locomotion.
- Highlight the significance of understanding tail power in relation to the overall movement and survival of fish in their natural habitats.
- Discuss what students will learn in the next lesson related to adaptations in aquatic animals.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook

**Learning Activities:**

- Summation of key points
- Preview of next lesson

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:**4

**TERM:**3

**SUBJECT:** BIOLOGY

**TOPIC:** SUPPORT & MOVEMENT IN PLANTS AND ANIMALS

**SUB TOPIC:** SUPPORT AND MOVEMENT IN MAMMALS. THE SKULL AND RIB CAGE.

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe the structure of the skull and rib cage.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students warmly and introduce the topic of support and movement in mammals, specifically focusing on the skull and rib cage.
- Briefly explain the significance of these structures in protecting vital organs and providing structural support.

**Resources:**

- Human skull
- Rib cage of rat/rabbit
- Textbook (KLB BK IV, pp. 119-120)

**Learning Activities:**

- Observation of human skull and rib cage of rat/rabbit
- Discussion on structure and function

**Lesson Content (30 minutes):**

*1. Understanding the Skull (15 minutes)*

- Discuss the human skull: its major parts (frontal bone, parietal bones, occipital bone, temporal bones, maxilla, mandible) and their functions in protection and support.
- Use the human skull model for a hands-on inspection. Encourage students to identify each part as you discuss.
- Compare the human skull with the skull of a rat/rabbit, emphasizing differences and similarities in structure and function.
- Key Activity: Group students into pairs for a brief discussion on the importance of the skull in daily life (e.g., protection of the brain, support for facial structure).

## *2. Understanding the Rib Cage (15 minutes)*

- Discuss the structure of the rib cage, including the rib bones, sternum, and thoracic vertebrae. Explain the role of the rib cage in protecting the heart and lungs.
- Use the rib cage of a rat/rabbit for a close inspection, pointing out adaptations that may differ between species.
- Key Activity: In small groups, have students briefly explore what might happen if the rib cage were injured (leading questions can guide this discussion).

### **Resources:**

- Human skull
- Rib cage of rat/rabbit
- KLB BK IV (pp. 119-120)

### **Learning Activities:**

- Hands-on observations of the skull and rib cage
- Group discussions on structure and function
- Analyzing adaptations between species

### **Conclusion (5 minutes):**

- Summarize the major points discussed: structure and function of the skull and rib cage, their roles in supporting and protecting vital organs. Highlight the importance of these structures in mammals' overall movement and stability.
- Prepare the students for the next lesson by previewing the upcoming topic related to locomotion in mammals.

### **Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, pp. 119-120)

### **Learning Activities:**

- Recap discussion
- Question-and-answer session to clarify any concepts

### **Teacher Self-Evaluation:**



**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

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**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: THE VERTEBRAL COLUMN. CERVICAL AND THORACIC VERTEBRAE.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

### **Objectives:**

By the end of the lesson, the learner should be able to:

1. Describe the features of the vertebral column.
2. Identify types of vertebrae.
3. Explain adaptations of cervical and thoracic vertebrae to their functions.

### **LESSON PRESENTATION**

#### **Introduction (5 minutes):**

- Begin with a warm greeting to the students.
- Introduce the topic of the vertebral column, explaining its importance in human anatomy and function.
- Outline the lesson objectives so students know what they will be learning.

#### **Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook: KLB BK IV. PP 121-2

#### **Learning Activities:**

- Brief discussion on the importance of the vertebral column in support and movement.

#### **Lesson Content (30 minutes):**

*Start with a detailed discussion of the vertebral column, its general definition, and key features such as:*

- Number of vertebrae (cervical, thoracic, lumbar, sacral, and coccygeal).
- Distinctive features like the vertebral body, arch, and processes.
- Move on to specifically discuss the cervical and thoracic vertebrae, emphasizing:
  - The unique characteristics of cervical vertebrae (e.g., atlas and axis, smaller size, flexibility).
  - The features of thoracic vertebrae (e.g., larger size, rib articulation, movement limitations).
- Discuss the functional adaptations of cervical vertebrae in supporting the head and allowing a wide range of motion.

- Explain the adaptations of thoracic vertebrae that support the rib cage and protect the thoracic organs while allowing some movement.
- Organize students into small groups to examine models or illustrations of cervical and thoracic vertebrae, identifying and discussing the different features and adaptations.

**Resources:**

- Models or illustrations of cervical and thoracic vertebrae
- KLB BK IV. PP 121-2 for reference
- Chalk board/white board for drawings/notes

**Learning Activities:**

- Group examination of vertebral models or illustrations.
- Group discussions focusing on identifying vertebrae and their adaptations.

**Conclusion (5 minutes):**

- Summarize the key points covered in the lesson: the features of the vertebral column, the types of vertebrae, and their specific adaptations.
- Prepare students for the next lesson by providing hints about the related topics that will be discussed.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook: KLB BK IV. PP 121-2

**Learning Activities:**

- Quick questions for students to answer about cervical and thoracic vertebrae to reinforce learning.

**Teacher Self-Evaluation:**

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**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: THE VERTEBRAL COLUMN. CERVICAL AND THORACIC VERTEBRAE.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

### **Objectives:**

By the end of the lesson, the learner should be able to:

1. Describe the features of the vertebral column.
2. Identify types of vertebrae.
3. Explain adaptations of cervical and thoracic vertebrae to their functions.

### **LESSON PRESENTATION**

#### **Introduction (5 minutes):**

Begin the lesson with a warm greeting to create a positive learning environment. Introduce the topic of the vertebral column and its significance in the skeletal system of vertebrates. Highlight the focus on cervical and thoracic vertebrae in this lesson.

#### **Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, PP 121-122)

#### **Learning Activities:**

- Brief discussion on the importance of the vertebral column.
- Overview of cervical and thoracic vertebrae.

#### **Lesson Content (30 minutes):**

##### *1. Discussion of Vertebral Column Features (10 minutes):*

- Explain the structure of the vertebral column, including its regions (cervical, thoracic, lumbar, sacral, and coccygeal).
- Describe the general features of vertebrae: vertebral body, neural arch, spinous process, transverse processes, and intervertebral discs.

##### *2. Identification of Types of Vertebrae (10 minutes):*

- Focus on cervical (7 vertebrae) and thoracic (12 vertebrae) vertebrae characteristics.
- Highlight key features such as:

- Cervical vertebrae: transverse foramen, bifid spinous processes.
- Thoracic vertebrae: costal facets for rib articulation, longer spinous processes.
- Use visual aids (diagrams, models) to enhance understanding.

### *3. Adaptations to Function (10 minutes):*

- Discuss the functional significance of cervical and thoracic vertebrae:
- Cervical vertebrae facilitate head movement and support the skull.
- Thoracic vertebrae provide attachment points for ribs and protect the thoracic organs, while allowing for limited movement.
- Engage students in an activity of examining real or model cervical and thoracic vertebrae, asking them to identify features and discuss their functions in pairs.

### **Resources:**

- Diagrams/models of cervical and thoracic vertebrae
- Textbook (KLB BK IV, PP 121-122)
- Real specimens (if available)

### **Learning Activities:**

- Group discussion on features and adaptations.
- Hands-on examination of vertebrae (real or model).

### **Conclusion (5 minutes):**

Summarize the salient points of the lesson, reiterating the features of the cervical and thoracic vertebrae and their functional adaptations. Encourage students to reflect on how these adaptations are crucial for the overall movement and support of the body.

### **Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, PP 121-122)

### **Learning Activities:**

- Brief recap and Q&A session to clarify any doubts or reinforce learning.
- Prepare students for the next lesson on the lumbar and sacral vertebrae.

### **Teacher Self-Evaluation:**

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SCHOOL/INSTITUTION: \_\_\_\_\_  
FORM:4  
TERM:3  
SUBJECT: BIOLOGY  
TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS  
SUB TOPIC: THORACIC AND LUMBAR VERTEBRAE.  
WEEK: \_\_\_\_\_ LESSON NO: \_\_\_\_\_  
DATE \_\_\_\_\_ TIME \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Explain adaptations of Thoracic and lumbar vertebrae to their functions.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students and introduce the topic of the day, explaining the significance of studying the thoracic and lumbar vertebrae in relation to support and movement in mammals.
- Ask a few questions to engage the students, such as "What do you think makes the thoracic vertebrae different from the lumbar vertebrae?"
- Emphasize the importance of these vertebral regions in terms of structural support for the body and protection for the spinal cord.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, pp. 122-123)

**Learning Activities:**

- Brief introduction to the topic
- Interactive question session

**Lesson Content (30 minutes):**

*1. Discussion on Thoracic Vertebrae (15 minutes):*

- Explain the structure of thoracic vertebrae and their typical features, such as the presence of rib articulations, which support rib cage formation.
- Discuss their role in protecting vital organs in the thoracic cavity and the importance of their attachment to the ribs for upper body support and movement.
- Use diagrams and models (if available) to illustrate these points.

*2. Discussion on Lumbar Vertebrae (15 minutes):*

- Explain the structure of lumbar vertebrae, focusing on their larger size and weight-bearing properties, which allow them to support the body's weight during various activities.
- Discuss adaptations such as the shape of the lumbar vertebrae that provide a wider range of motion and better posture support in humans.
- Encourage students to compare and contrast the features of thoracic and lumbar vertebrae.

**Resources:**

- KLB BK IV textbook (pp. 122-123)
- Diagrams of thoracic and lumbar vertebrae
- Models of vertebrae (if accessible)

**Learning Activities:**

- Group discussions on vertebrae features
- Drawing labeled diagrams of thoracic and lumbar vertebrae
- Interactive comparisons between the two vertebral regions

**Conclusion (5 minutes):**

- Summarize the critical features and functions of thoracic and lumbar vertebrae, highlighting their adaptations for specific roles in the skeletal system.
- Reinforce the significance of understanding these adaptations in the context of human anatomy and physiology.
- Prepare students for the next lesson, which may cover related topics on the spinal column or movement in other organisms.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV)

**Learning Activities:**

- Summary of key points
- Discussion on the transition to the next lesson

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: THORACIC AND LUMBAR VERTEBRAE.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

Explain adaptations of thoracic and lumbar vertebrae to their functions.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson by greeting the students warmly. Introduce the topic of thoracic and lumbar vertebrae by explaining their significance in the human skeleton. Discuss briefly how these vertebrae support vital functions such as movement and protection of the spinal cord.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook: KLB BK IV, PP 122-3

**Learning Activities:**

- Brief discussion on the importance of the spinal column.
- Introduction of the thoracic and lumbar regions in the vertebral column.

**Lesson Content (30 minutes):**

In this section, start with a detailed overview of the thoracic and lumbar vertebrae, highlighting their anatomical features, differences, and their functional adaptations. Facilitate a discussion on the following key points:

- Structure of thoracic vertebrae: emphasize the presence of rib articulations which support the thoracic cage.
- Functionality of thoracic vertebrae: discuss how they allow rotation and some lateral movement.
- Structure of lumbar vertebrae: highlight the larger body size for increased weight-bearing capabilities.
- Functionality of lumbar vertebrae: focus on stability, support for the lower back, and resistance to torsion.

Group students to examine physical models or illustrations of thoracic and lumbar vertebrae. Encourage them to draw labeled diagrams to reinforce learning visually.

Conduct a brief discussion afterwards to clarify doubts and encourage peer teaching, fostering a deeper understanding of the adaptations of these vertebrae to their specific functions.

**Resources:**

- Physical models of thoracic and lumbar vertebrae (if available)
- Illustrations and diagrams from KLB BK IV, PP 122-3
- Chalk board/white board

**Learning Activities:**

- Group examination of vertebrae models.
- Drawing labeled diagrams.
- Discussion about adaptations and functions of thoracic and lumbar vertebrae.

**Conclusion (5 minutes):**

Conclude by summarizing the key points discussed concerning the anatomical features and functional adaptations of the thoracic and lumbar vertebrae. Highlight the unique characteristics and importance of both types in terms of support and movement in the human body. Prepare students for the next lesson by introducing a brief overview of the next topic related to vertebral functions.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook: KLB BK IV, PP 122-3

**Learning Activities:**

- Summarizing discussion points.
- Q&A session to clarify remaining doubts.

**Teacher Self-Evaluation:**



**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: THORACIC AND LUMBAR VERTEBRAE.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

1. Explain adaptations of Thoracic and lumbar vertebrae to their functions.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Begin by greeting the students and introducing the topic of Thoracic and Lumbar Vertebrae.
- Explain the significance of studying these vertebrae in relation to support and movement within the human body, linking it to overall anatomy.
- Conclude with a brief overview of what will be covered in the lesson.

**Resources:**

Chalk board/white board  
A piece of chalk/marker pen  
Textbook (KLB BK IV, pp. 122-123)

**Learning Activities:**

Discussion on the importance of vertebrae in the human body.

**Lesson Content (30 minutes):**

*Detailed Exploration (10 minutes):*

- Provide an engaging explanation of the thoracic and lumbar vertebrae, highlighting their structures and locations within the vertebral column.
- Discuss the specific adaptations of the thoracic vertebrae that allow for rib attachment and protection of the heart and lungs. Focus on their stability and flexibility for respiration.
- Explain the lumbar vertebrae's adaptations that facilitate weight-bearing and bipedal locomotion, discussing their larger size for load distribution.

*Diagrams and Models (10 minutes):*

- Guide the students to examine thoracic and lumbar vertebrae through labeled diagrams.
- Encourage them to draw their own diagrams while labeling the main features. Use a model if available to enhance understanding.

- Explain the orientation and differences between the thoracic and lumbar vertebrae with the help of visual aids.

*Group Discussion (10 minutes):*

- Facilitate a brief discussion by dividing students into small groups to share their drawings and discuss key differences between thoracic and lumbar vertebrae.
- Each group will summarize their findings, focusing on how the structure relates to function in terms of mobility and support.

**Resources:**

Textbook (KLB BK IV, pp. 122-123)

Diagrams and anatomical models (if available)

**Learning Activities:**

Examining vertebrae through models and diagrams.

Drawing labeled diagrams of thoracic and lumbar vertebrae.

Group discussions summarizing key points.

**Conclusion (5 minutes):**

- Recap the key points discussed, emphasizing how the unique adaptations of thoracic and lumbar vertebrae serve their specific functions in the body.
- Prepare students for the next lesson, hinting at related topics such as spinal health or other components of the skeletal system.
- Encourage students to review the diagrams and information covered.

**Resources:**

Chalk board/white board

A piece of chalk/marker pen

Textbook (KLB BK IV, pp. 122-123)

**Learning Activities:**

Summary discussion of key points.

Encouragement to further study and review.

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: THORACIC AND LUMBAR VERTEBRAE.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

### **Objectives:**

By the end of the lesson, the learner should be able to:

- Explain adaptations of Thoracic and Lumbar vertebrae to their functions.

### **LESSON PRESENTATION**

#### **Introduction (5 minutes):**

- Greet the students warmly to create a conducive learning environment.
- Introduce the topic of Thoracic and Lumbar vertebrae, explaining their importance within the human skeletal system and their roles in support and movement.
- Engage students by asking if anyone has encountered issues related to the back or spine, introducing a relatable context for the lesson.

#### **Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 122-3)

#### **Learning Activities:**

- Brief discussion on students' prior knowledge.
- Introduction to the vertebral column and its sections.

#### **Lesson Content (30 minutes):**

- Begin with a visual examination of the thoracic and lumbar vertebrae using models or diagrams. Discuss their anatomical locations, shapes, and sizes.
- Explain the adaptations of the thoracic vertebrae (e.g., attachment for ribs, support for the thoracic cavity) and lumbar vertebrae (e.g., larger and stronger structure for weight-bearing and movement).
- Divide students into small groups and assign each group a specific vertebra to examine. Each group will draw labeled diagrams representing the vertebra's features and adaptations.
- Encourage a brief group discussion after presentations to share insights and compare differences between thoracic and lumbar adaptations.

**Resources:**

- Models of thoracic and lumbar vertebrae
- Textbooks (KLB BK IV)
- Art supplies (paper, pencils) for drawing diagrams

**Learning Activities:**

- Visual examination of vertebrae models.
- Group work for drawing labeled diagrams.
- Presentations and discussions among groups.

**Conclusion (5 minutes):**

- Recap the main adaptations of thoracic and lumbar vertebrae and their functional significance.
- Highlight the importance of understanding these adaptations for health and physical activities.
- Prepare students for the next lesson by hinting at related topics such as spinal health or more about the vertebral column.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 122-3)

**Learning Activities:**

- Summary of key points.
- Q&A session to reinforce understanding.

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: THORACIC AND LUMBAR VERTEBRAE.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Explain adaptations of Thoracic and lumbar vertebrae to their functions.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Start the lesson with a warm greeting to the students to create a positive learning environment. Introduce the topic of the thoracic and lumbar vertebrae, emphasizing their role in the skeletal system of mammals. Explain how these vertebrae contribute to support and movement.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 122-3)

**Learning Activities:**

- Quick questioning to assess prior knowledge
- Listing of skeletal components in students' notes

**Lesson Content (30 minutes):**

*1. Discussion of Thoracic Vertebrae (15 minutes):*

- Describe the structure, function, and adaptations of the thoracic vertebrae, which articulate with the ribs and provide support for the upper body.
- Discuss their role in protecting the thoracic organs and supporting respiratory mechanics.
- Highlight specific adaptations such as the spinous processes which are downward sloping to allow for muscle attachment and limit the movement to ensure stability.

*2. Discussion of Lumbar Vertebrae (15 minutes):*

- Explore the structure, function, and adaptations of the lumbar vertebrae, which are larger and bear more weight compared to thoracic vertebrae.
- Explain their role in supporting the lower back during activities and their importance in flexibility and movement.

- Discuss adaptations such as the wider intervertebral discs and robust transverse processes that facilitate stability while allowing for a range of motion.

**Resources:**

- KLB BK IV. PP 122-3
- Visual aids (diagrams of vertebrae, skeletal models if available)

**Learning Activities:**

- Group discussion and comparison of thoracic vs. lumbar vertebrae
- Drawing and labeling diagrams of both vertebrae
- Questioning to ensure understanding and retention

**Conclusion (5 minutes):**

Recap the key points regarding the adaptations of thoracic and lumbar vertebrae and their functions. Reinforce the connection between vertebral structure and its purpose in biomechanics. Encourage students to prepare questions for further exploration in the next lesson, which may focus on the whole vertebral column or related anatomical structures.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 122-3)

**Learning Activities:**

- Summary discussion led by students
- Preparation for next lesson by brainstorming questions

**Teacher Self-Evaluation:**

TEACHER'S NAME: \_\_\_\_\_ TSC NO: \_\_\_\_\_  
SCHOOL/INSTITUTION: \_\_\_\_\_  
FORM:4  
TERM:3  
SUBJECT: BIOLOGY  
TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS  
SUB TOPIC: THE SACRAL AND CAUDAL VERTEBRAE.  
WEEK: \_\_\_\_\_ LESSON NO: \_\_\_\_\_  
DATE \_\_\_\_\_ TIME \_\_\_\_\_

### Objectives:

By the end of the lesson, the learner should be able to:

1. Describe the features of the sacral and caudal vertebrae.
2. Explain adaptations of sacral and caudal vertebrae to their functions.

### LESSON PRESENTATION

#### Introduction (5 minutes):

Begin the lesson with a warm greeting to create a positive learning environment. Introduce the topic of sacral and caudal vertebrae, explaining their importance in the skeletal system of vertebrates. Mention how these vertebrae are adapted for specific functions like support and movement.

#### Resources:

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, p. 124)

#### Learning Activities:

- Brief introductory discussion about vertebrae.
- Checking students' prior knowledge.

#### Lesson Content (30 minutes):

##### *1. Exploration of the sacral vertebrae:*

- Discuss the structure of the sacral vertebrae, emphasizing their fusion to form the sacrum and their role in supporting the pelvis and providing stability.
- Highlight key features such as the dorsal sacral foramina and the auricular surfaces.

##### *2. Exploration of the caudal vertebrae:*

- Describe the anatomy of caudal vertebrae, particularly in different vertebrate species, illustrating how they vary in number and shape.
- Discuss functions such as balance and propulsion in animals like fish and certain mammals.

### *3. Adaptations:*

- Divide students into small groups to examine how the sacral vertebrae have adapted for bipedalism in humans and how caudal vertebrae adapt in different animals (e.g., tail in kangaroos for balance).
- Engage in a hands-on examination of models or diagrams of these vertebrae and discuss their findings with the class.

Conclude this section by emphasizing how the structure of these vertebrae is intrinsically linked to their function in the body.

#### **Resources:**

- Textbook (KLB BK IV, p. 124)
- Models or diagrams of vertebrae

#### **Learning Activities:**

- Group exploration of models/diagrams.
- Class discussion on the features and adaptations.
- Drawing labeled diagrams of sacral and caudal vertebrae.

#### **Conclusion (5 minutes):**

Summarize the key features of the sacral and caudal vertebrae covered in the lesson. Reinforce the idea of structure-function relationships in biological systems. Prepare students for the next lesson by hinting at the next topic in skeletal systems or vertebrate anatomy.

#### **Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, p. 124)

#### **Learning Activities:**

- Summary discussion.
- Quick review questions to assess understanding.

#### **Teacher Self-Evaluation:**



**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: THE SACRAL AND CAUDAL VERTEBRAE.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

1. Describe the features of the sacral and caudal vertebrae.
2. Explain adaptations of sacral and caudal vertebrae to their functions.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Begin with a warm greeting to the students to create a welcoming environment.
- Introduce the topic by explaining the importance of the sacral and caudal vertebrae in the context of support and movement in vertebrates.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. P 124)

**Learning Activities:**

- Brief discussion
- Reading from the textbook

**Lesson Content (30 minutes):**

- Begin with an examination of the sacral vertebrae, highlighting their structure and functions, such as their role in providing stability and support to the pelvic girdle. Discuss how these vertebrae are fused into a single sacrum in most mammals.
- Proceed to the caudal vertebrae, describing their features and variability among different species, including their role in balance and mobility, particularly in animals with long tails.
- Engage students in a hands-on activity where they examine models or images of the sacral and caudal vertebrae.
- Have students draw and label diagrams of both the sacral and caudal vertebrae, emphasizing the different sections and their respective functions.
- Facilitate a brief group discussion that encourages students to share their thoughts on the adaptation of these vertebrae for various environmental challenges and locomotion methods.

- Conclude this section with a review of the key features and adaptations of both the sacral and caudal vertebrae.

**Resources:**

- Models/images of sacral and caudal vertebrae
- Textbook (KLB BK IV. P 124)
- Diagramming materials (paper, colored pencils)

**Learning Activities:**

- Examination of vertebrae models/images
- Drawing labeled diagrams
- Group discussion

**Conclusion (5 minutes):**

- Summarize the key points discussed during the lesson, reiterating the importance of the sacral and caudal vertebrae in relation to movement and support in animals.
- Prepare students for the next lesson by hinting at the upcoming topics related to the vertebral column and its functions.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. P 124)

**Learning Activities:**

- Summary discussion
- Preparation for the next lesson

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: THE APPENDICULAR SKELETON.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe the features of the appendicular skeleton.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Greet the students warmly and introduce the topic of the appendicular skeleton. Explain that the appendicular skeleton includes the bones of the limbs and girdles that attach them to the axial skeleton.

- State that today, they will examine the appendicular skeleton of a rabbit or sheep to enhance their understanding.

**Resources:**

- Appendicular skeleton of a rabbit/sheep
- Chalk board/white board
- A piece of chalk/marker pen
- KLB BK IV. PP 124-5

**Learning Activities:**

- Examination of the appendicular skeleton
- Brief discussion

**Lesson Content (30 minutes):**

Engage the students in a detailed discussion of the appendicular skeleton.

- Start by explaining the key components of the appendicular skeleton, including the pectoral (shoulder) girdle, pelvic (hip) girdle, and the bones of the limbs.
- Ask students to participate in the examination of the appendicular skeleton of the rabbit/sheep, guiding them to identify and describe specific features such as:
  - Scapula, clavicle (pectoral girdle)
  - Humerus, radius, ulna (forelimb bones)
  - Carpals, metacarpals, phalanges (hand bones)
  - Pelvis (pelvic girdle)

- Femur, patella, tibia, fibula (hind limb bones)
- Tarsals, metatarsals, phalanges (foot bones)

Encourage students to work in pairs or small groups to discuss the functional significance of each bone and its role in support, movement, and locomotion. Circulate the room to facilitate discussions and provide guidance as needed.

- Conclude this section with a summarization of the distinct features of the appendicular skeleton while reiterating the importance of these bones in movement and support.

**Resources:**

- Appendicular skeleton of a rabbit/sheep
- KLB BK IV. PP 124-5
- Chalk board/white board
- A piece of chalk/marker pen

**Learning Activities:**

- Group examination of the appendicular skeleton
- Pair discussions on the functional significance of each bone
- Recap on the key features of the appendicular skeleton

**Conclusion (5 minutes):**

Summarize the key points from the lesson. Encourage students to remember the names and functions of various bones within the appendicular skeleton and their importance in animal movement. Remind students to review Chapter on the appendicular skeleton in their textbooks for deeper understanding and prepare for the next lesson, which will cover the axial skeleton.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 124-5)

**Learning Activities:**

- Recap of the lesson
- Discussion on the importance of the appendicular skeleton in animal movement

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: THE APPENDICULAR SKELETON.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe the features of the appendicular skeleton.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson with a friendly greeting, setting a positive tone for the class. Introduce the topic of the appendicular skeleton, connecting it to the previous lesson on the axial skeleton to create continuity. Highlight how the appendicular skeleton is crucial for movement and support in animals.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Brief discussion on the importance of the appendicular skeleton in movement.

**Lesson Content (30 minutes):**

*1. Discussion of the Appendicular Skeleton (15 minutes):*

- Define what the appendicular skeleton is and its components: limbs (upper and lower), pectoral girdle, and pelvic girdle.
- Explain the function of each part and how they contribute to movement in animals.

*2. Examination of the Appendicular Skeleton (15 minutes):*

- Organize students into small groups and provide each group with either the appendicular skeleton of a rabbit or a sheep.
- Instruct students to observe and identify the major bones (e.g., humerus, radius, ulna, femur, etc.) and the girdles. Encourage them to discuss the characteristics of these bones and how they relate to the animal's function and movement.

Conclude this section by allowing students to share their observations with the class, reinforcing their understanding through peer learning.

**Resources:**

- Appendicular skeleton of a rabbit/sheep
- KLB BK IV, pp. 124-125

**Learning Activities:**

- Group discussions
- Hands-on examination of the appendicular skeleton models

**Conclusion (5 minutes):**

Summarize the key features of the appendicular skeleton discussed in the lesson. Emphasize its role in support and movement, and how understanding its anatomy is vital for comprehending larger biological concepts. Prepare students for the next lesson by indicating that they will explore the musculature associated with the appendicular skeleton.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Recap of the key points discussed
- Brief Q&A session to clarify any misunderstandings

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: THE APPENDICULAR SKELETON.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe the features of the appendicular skeleton.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students and introduce the topic of the appendicular skeleton.
- Pose a question to stimulate interest: "What do you think the appendicular skeleton includes?"
- Explain that today's lesson will focus on the features and significance of the appendicular skeleton in mammals.

**Resources:**

- Chalk board/White board
- A piece of Chalk/Marker pen
- Textbook (KLB BK IV, pp. 124-125)

**Learning Activities:**

- Question and answer session.
- Brief student-led discussion on prior knowledge of the skeletal system.

**Lesson Content (30 minutes):**

- Begin by defining the appendicular skeleton and its role in support and movement, emphasizing its components: limbs and girdles.
- Discuss the two main parts:
- Pectoral Girdle: Explain the features and function, focusing on the scapula and clavicle.
- Pelvic Girdle: Describe its structure (pubis, ischium, and ilium) and function in locomotion and support.
- Use the appendicular skeleton of the rabbit/sheep as a hands-on resource.
- Group Activity: Divide the class into small groups and have them examine the appendicular skeleton. Ask them to identify features such as the femur, tibia, humerus, and radius.
- Facilitate a brief discussion within groups about the differences they observe compared to the axial skeleton.

- Reconvene as a class and have each group share their findings, fostering a deeper understanding through peer interaction.

**Resources:**

- Appendicular skeleton models (rabbit/sheep)
- KLB BK IV (pp. 124-125)

**Learning Activities:**

- Group examination of the appendicular skeleton.
- Presentation of group findings to the class.
- Teacher-led class discussion on the significance of these skeletal features.

**Conclusion (5 minutes):**

- Summarize the key points discussed: the function of the appendicular skeleton, its major components, and their relevance to movement.
- Prepare students for the next lesson by hinting at the comparison between the appendicular and axial skeletons and the overall integration of these systems in mammals.

**Resources:**

- Chalk board/White board
- A piece of Chalk/Marker pen
- Textbook (KLB BK IV, pp. 124-125)

**Learning Activities:**

- Quick recap of the day's lesson through a question-and-answer format.
- Assign students to read the relevant sections in their textbooks for further understanding of the appendicular skeleton.

**Teacher Self-Evaluation:**



**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: FORE LIMB.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify bones of the fore limb.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students warmly to create a positive learning environment.
- Introduce the topic of fore limb anatomy, highlighting its importance in support and movement.
- Mention that today's focus will include identifying the bones that comprise the fore limb.

**Resources:**

- Bones of the fore limb
- Chalk board/white board
- A piece of chalk/marker pen
- KLB BK IV. PP 126-7 (Textbook)

**Learning Activities:**

- Discussion about fore limb anatomy
- Examination of bones
- Drawing labeled diagrams

**Lesson Content (30 minutes):**

- Begin with a detailed discussion on the fore limb, explaining its function and significance for different animals (e.g., humans, birds). Highlight how the structure of the fore limb can vary widely among species depending on their environment and lifestyle.

*Introduce the major bones that constitute the fore limb:*

- Humerus
- Radius
- Ulna
- Carpals
- Metacarpals
- Phalanges

- Organize the students into small groups to examine the actual bones (if available) or pictures/diagrams. Encourage them to discuss the features of each bone and their respective locations in the fore limb.
- Facilitate hands-on activities where students sketch the fore limb and label the bones they discussed.
- Once sketches are completed, hold a class discussion where groups share their diagrams and observations. Correct any misconceptions and reinforce learning by asking guiding questions about how the bones work together to allow movement.

**Resources:**

- Bones of the fore limb
- KLB BK IV. PP 126-7 (Textbook)
- Various diagrams/pictures of fore limbs (printed or digital)

**Learning Activities:**

- Group examination of bones/pictures
- Sketching and labeling diagrams
- Class discussion on findings

**Conclusion (5 minutes):**

- Recap the main points discussed about the bones of the fore limb, emphasizing the importance of each in terms of movement and function.
- Encourage students to review the material covered for the next lesson, where they will explore the relationships between different bones and muscles in the fore limb.
- Thank the students for their participation and enthusiasm.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- KLB BK IV. PP 126-7 (Textbook)

**Learning Activities:**

- Summary of key points
- Discussion of next lesson's focus

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: FORE LIMB.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify bones of the fore limb.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson by greeting the students and explaining the significance of the fore limb in support and movement among animals. Introduce the specific bones that will be covered in today's lesson.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook: KLB BK IV, pp. 126-7

**Learning Activities:**

- Introductory discussion
- Explanation of resources

**Lesson Content (30 minutes):**

*1. Examine Bones of the Fore Limb (15 minutes):*

Provide students with real or model bones of the fore limb. Allow them to explore and handle the bones. Ask guiding questions to help them identify each bone and its role. The bones to be covered include:

- Humerus
- Radius
- Ulna
- Carpals
- Metacarpals
- Phalanges

*2. Drawing Labeled Diagrams (10 minutes):*

Instruct students to draw a labeled diagram of the fore limb on their worksheets. They should include all the bones they identified during the examination. Encourage creativity and accuracy.

*3. Group Discussion (5 minutes):*

Facilitate a short group discussion where students can share what they learned about the bones of the fore limb. Encourage them to ask questions and clarify any doubts.

**Resources:**

- Real or model bones of the fore limb
- Worksheets for drawing
- Textbook: KLB BK IV, pp. 126-7

**Learning Activities:**

- Hands-on examination of bones
- Diagram drawing exercise
- Group discussion

**Conclusion (5 minutes):**

Summarize the lesson by briefly reviewing the bones of the fore limb discussed and their functions. Ask a few students to share one thing they learned today. Prepare the students for the next lesson which will build on this knowledge.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook: KLB BK IV, pp. 126-7

**Learning Activities:**

- Summary discussion
- Student shares

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: FORE LIMB.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE:** \_\_\_\_\_ **TIME:** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify bones of the fore limb.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students and introduce the topic of 'Fore Limb' by discussing its significance in locomotion and support in animals. Explain that today's focus will be on learning about the different bones that make up the fore limb.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV)

**Learning Activities:**

- Brief discussion
- Overview of resources and activities

**Lesson Content (30 minutes):**

- Begin by displaying an actual model or images of the fore limb bones (e.g., humerus, radius, ulna, carpals, metacarpals, and phalanges) to provide a visual understanding.
- Divide the class into small groups. Provide each group with a set of replica bones if available or use printed diagrams from the textbook to examine the bones of the fore limb closely.
- Instruct each group to identify and name each bone, discussing its location and function. Encourage them to note any interconnections or unique features.
- Afterwards, bring the class back together for a class discussion, where groups can share their findings.
- As groups present, students will draw labeled diagrams of the fore limb bones in their notebooks, reinforcing their learning visually.
- Conclude this segment by asking questions to assess understanding and clarity, ensuring every student is engaged.

**Resources:**

- Bones of the fore limb (models or replicas)
- KLB BK IV (pages 126-127)
- Diagrams/printed images of fore limb bones

**Learning Activities:**

- Group examination of bones
- Group discussions
- Drawing labeled diagrams

**Conclusion (5 minutes):**

- Recap the key bones of the fore limb discussed during the lesson, emphasizing their names, locations, and functions. Stress the importance of these bones in animal movement and support.
- Prepare students for the next lesson by hinting at what will come next regarding movement in animals.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV)

**Learning Activities:**

- Summary discussion
- Q&A session

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: HIND LIMB.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify bones of the hind limb.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Begin the lesson with a warm greeting to establish an inviting atmosphere for learning.
- Introduce the topic of the hind limb, explaining its significance in support and movement in animals.
- Set the context by highlighting the relationship between the structure of the hind limb and its function in locomotion.
- Gradually build interest by asking the students about their prior knowledge of the anatomy of limbs.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, P 127)
- Bones of the hind limb

**Learning Activities:**

- Class discussion on prior knowledge
- Introduction to the topic

**Lesson Content (30 minutes):**

- Begin with a presentation summarizing the components of the hind limb, detailing the major bones (femur, tibia, fibula, patella, etc.).
- Distribute physical models or drawings of the hind limb bones for students to examine closely.
- Engage students in a hands-on activity where they can hold and identify each bone, discussing its shape, structure, and role in locomotion.
- Conduct a group drawing activity where students create labeled diagrams of the hind limb on large paper.

- Facilitate a class discussion, encouraging students to share their findings and insights from the hands-on examination and drawing activities.
- Highlight and explain the relationship between the bones discussed and their functions in movement.
- Conclude with a recap of identified bones and their functions in a structured format.

**Resources:**

- Bones of the hind limb
- KLB BK IV, P 127

**Learning Activities:**

- Hands-on examination of bones
- Group drawing and labeling of hind limb
- Class discussion

**Conclusion (5 minutes):**

- Summarize the key points covered during the lesson, especially focusing on the names and functions of the hind limb bones.
- Emphasize the importance of understanding limb anatomy in the study of biology and its applications in real-life scenarios.
- Prepare students for the next lesson by hinting at the upcoming discussion on the forelimbs and their comparative anatomy.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV)

**Learning Activities:**

- Summary discussion
- Preparation for the next topic

**Teacher Self-Evaluation:**



**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: HIND LIMB.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify bones of the hind limb.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students warmly and engage them with a question about the importance of the hind limb in animals.
- Briefly introduce the significance of studying hind limb anatomy in understanding the movement and support in animals.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. P 127)

**Learning Activities:**

- Questioning session to engage students
- Introduction to resources

**Lesson Content (30 minutes):**

- Begin by displaying the bones of the hind limb. Explain each bone's name and location, including the femur, tibia, fibula, patella, tarsals, metatarsals, and phalanges.
- Divide the class into small groups and provide each group with physical specimens or images of the bones for hands-on examination.
- Guide each group to discuss their findings, allowing them to compare and contrast the different bones and their functions.
- After the group discussion, have students individually draw labeled diagrams of the hind limb, showcasing all the bones studied.
- Facilitate a class discussion where students can share their diagrams and clarify any doubts.
- Reinforce the material by discussing common injuries and diseases related to the hind limb, emphasizing each bone's role in movement.

**Resources:**

- Bones of the hind limb
- KLB BK IV. P 127
- Images/diagrams of the hind limb for visual reference

**Learning Activities:**

- Hands-on examination of bones/images
- Group discussions
- Individual drawing of labeled diagrams
- Class discussion and sharing of diagrams

**Conclusion (5 minutes):**

- Summarize the key points covered in the lesson, including the names and functions of the bones in the hind limb.
- Highlight the importance of understanding hind limb anatomy not just for biology, but also for applications in health and veterinary sciences.
- Prepare students for the next lesson by introducing the upcoming topic related to the muscular system associated with the hind limb.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. P 127)

**Learning Activities:**

- Summary of key points
- Questions from students to clarify concepts
- Presentation of next lesson topic

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: MOVABLE JOINTS.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify features of movable joints.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students and create a welcoming environment for learning.
- Briefly introduce the topic of movable joints, highlighting their importance in human and animal movement.
- Encourage students to share any prior knowledge about joints and movements.

**Resources:**

- Synovial joint model
- Chalk board/white board
- Textbook (KLB BK IV P 127)

**Learning Activities:**

- Whole-class discussion
- Visual examination of the synovial joint model

**Lesson Content (30 minutes):**

*1. Examination of the Synovial Joint (15 minutes):*

- Use the synovial joint model to demonstrate the structure of movable joints.
- Explain the key features: articular cartilage, synovial membrane, joint capsule, ligaments, and synovial fluid.
- Encourage students to interact with the model, pointing out each feature as discussed.

*2. Brief Discussion (15 minutes):*

- Facilitate a discussion on the function of movable joints in various animals and how they adapt to different movements (e.g., walking, flying, swimming).
- Allow students to ask questions, providing clarifications and additional information as needed.

- Conclude by linking the structure of movable joints to their functional significance in daily life and sports.

**Resources:**

- Synovial joint model
- KLB BK IV P 127
- Chalk board/white board

**Learning Activities:**

- Hands-on examination of the synovial joint model
- Group discussions on the adaptations of movable joints in different species

**Conclusion (5 minutes):**

- Summarize the structural features of movable joints discussed, including the role of each component.
- Highlight the adaptability and importance of movable joints in facilitating various physical activities.
- Prepare students for the next lesson by giving a hint about the relationship between movable joints and muscle action.

**Resources:**

- Chalk board/white board
- Textbook (KLB BK IV P 127)

**Learning Activities:**

- Whole-class recap and Q&A

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: MOVABLE JOINTS.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify features of movable joints.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin by greeting the students and reinforcing the importance of understanding how joints function in movement. Introduce the topic of movable joints, highlighting their significance in the human skeleton and overall mobility. Mention that today's lesson will focus on synovial joints and their characteristics.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook (KLB BK IV)

**Learning Activities:**

- Brief discussion to engage students and assess prior knowledge.

**Lesson Content (30 minutes):**

*1. Examine the Synovial Joint (15 minutes):*

- Use the synovial joint model to provide a visual and tactile experience. Show students the key features, including the synovial capsule, synovial fluid, ligaments, and cartilage.
- Engage students by asking them to identify these features directly on the model, guiding them through the anatomy of a synovial joint.
- Encourage questions and promote a discussion on how these features contribute to the functionality of movable joints.

*2. Brief Discussion (15 minutes):*

- Facilitate a group discussion on the role of movable joints in human activities, injuries related to these joints (e.g., arthritis, dislocations), and the importance of joint health.

- Ask students to share examples of different types of movable joints (like hinge joints, ball-and-socket joints) and their specific functions.
- Summarize the discussion, connecting it with the textbook information to reinforce learning.

**Resources:**

- Synovial joint model
- KLB BK IV (P 127)

**Learning Activities:**

- Hands-on examination of the synovial joint model.
- Group discussion and sharing.

**Conclusion (5 minutes):**

Reiterate the key features of movable joints covered in the lesson. Highlight the importance of these joints in daily activities and the human body's overall mobility. Prepare students for the next lesson by hinting at the upcoming topics that will further explore joint movement and health.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook (KLB BK IV)

**Learning Activities:**

- Summary recap and Q&A session.

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:**4

**TERM:**3

**SUBJECT:** BIOLOGY

**TOPIC:** SUPPORT & MOVEMENT IN PLANTS AND ANIMALS

**SUB TOPIC:** BALL AND SOCKET JOINT.

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify features of ball and socket joint.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students and engage them in a brief discussion on what they understand about joints in the human body.
- Introduce the topic of ball and socket joints, explaining their importance in allowing a wide range of motion.
- Inform students that they will be examining a synovial joint model to observe and identify the features of this type of joint.

**Resources:**

- Synovial joint model
- Chalk board/white board
- Textbook (KLB BK IV, P 128)

**Learning Activities:**

- Group discussion on joint types
- Introduction to synovial joints

**Lesson Content (30 minutes):**

- Begin by presenting the synovial joint model to the class, directing students' attention to the ball and socket part of the model.
- Explain the key features of ball and socket joints such as:
  - Structure: The spherical head of one bone fits into the cup-like socket of another bone.
  - Movement: Allow for rotational movement (e.g., shoulder and hip joints).
  - Stability and flexibility: Discuss how these joints balance flexibility with stability.

- Organize the class into small groups and assign each group to examine specific parts of the synovial joint model.
- Each group should discuss their observations regarding:
  - The shape of the joint
  - Range of movement allowed
  - Comparison with other types of joints (e.g., hinge joints)
- After the group discussions, bring the class back together to share their findings. Encourage students to ask questions and clarify any misconceptions.

**Resources:**

- Synovial joint model
- KLB BK IV, P 128

**Learning Activities:**

- Hands-on examination of the model
- Group discussions and observations
- Class sharing on findings

**Conclusion (5 minutes):**

- Summarize the key points discussed during the lesson, reiterating the features of ball and socket joints and their significance in human movement.
- Encourage students to think about where they see these joints in action in their daily lives (e.g., throwing a ball, walking).
- Prepare them for the next lesson by hinting at the different forms of movement facilitated by various joint types.

**Resources:**

- Chalk board/white board
- Textbook (KLB BK IV, P 128)

**Learning Activities:**

- Summary discussion
- Preparation for next lesson

**Teacher Self-Evaluation:**



**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: THE HINGE JOINT. MOVEMENT OF A JOINT.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

1. Identify features of hinge joints.
2. Describe the movement of a joint.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students warmly to create a welcoming learning environment.
- Introduce the topic of hinge joints by explaining that they are a type of joint found in our body that allows movement in one direction, similar to the way a door swings open and shut.
- Pose a question to engage students, such as, "Can anyone tell me where we can find hinge joints in our body?"

**List resources used:**

- Chalk board/white board
- A piece of chalk/marker pen
- KLB BK IV.

**List learning activities:**

- Examine a hinge joint.
- Observe movement of the forearm.
- Discussion.

**Lesson Content (30 minutes):**

- Begin by explaining the structure of hinge joints, highlighting their key features such as the end of the bones being shaped to fit together allowing for only back-and-forth motion. Use illustrative diagrams to show examples of hinge joints in the human body, like the elbow and knee.
- Engage students in a group discussion about the differences between hinge joints and other types of joints (e.g. ball-and-socket joints). Encourage them to think about the mobility and range of motion in both types of joints.

- Demonstrate the movement of a hinge joint using your own arm. Ask students to follow along and observe how the forearm moves in one direction, showcasing flexion and extension.
- Divide students into pairs and have them examine a model of a hinge joint or perform a simple activity demonstrating hinge joint movement using their own bodies, such as bending and straightening their arms.
- Conclude this section by discussing the importance of hinge joints in everyday activities like lifting or throwing.

**List resources used:**

- Illustrative diagrams
- KLB BK IV. PP 128-9

**List learning activities:**

- Group discussion
- Hands-on examination of hinge joints
- Movement demonstration

**Conclusion (5 minutes):**

- Summarize the key points covered about hinge joints and their movements, reiterating their importance in our daily activities.
- Ask the students to share one new thing they learned about hinge joints.
- Prepare students for the next lesson by hinting at exploring another type of joint.

**List resources used:**

- Chalk board/white board
- A piece of chalk/marker pen
- KLB BK IV.

**List learning activities:**

- Student sharing
- Overview of next topic

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: THE HINGE JOINT. MOVEMENT OF A JOINT.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Identify features of hinge joints.
- Describe the movement of a joint.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students warmly and engage them with a brief overview of how joints play a crucial role in movement for both plants and animals.
- Explain the specific focus of today's lesson on hinge joints, demonstrating the importance of understanding different types of joints.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- KLB BK IV (pages 128-9)
- Illustrative diagrams

**Learning Activities:**

- Initial discussion about joints
- Interactive questioning to gauge prior knowledge

**Lesson Content (30 minutes):**

*Discussion on Features of Hinge Joints:*

Define hinge joints and explain their characteristics, such as:

- Allowing movement in one plane (flexion and extension).
- Specific examples (e.g., elbow and knee joints).
- Use visual aids, such as illustrative diagrams to show the structure of hinge joints.

*Movement of a Joint:*

Describe how hinge joints enable movement:

- Explain flexion and extension.

- Show how the forearm moves at the elbow as a practical demonstration.
- Have students observe and engage in movements by extending and bending their own arms; encouraging them to feel how the hinge joint functions.

*Group Activity:*

Divide the class into small groups and provide each group with diagrams and reference materials.

- Ask them to discuss further characteristics and movements associated with hinge joints and prepare to share with the class.

Conclude this section by summarizing the key points discussed and how they relate to the functions of hinge joints.

**Resources:**

- KLB BK IV (pages 128-9)
- Illustrative diagrams

**Learning Activities:**

- Group discussion about hinge joints
- Practical demonstration of arm movements
- Group analysis of provided diagrams

**Conclusion (5 minutes):**

- Recap the key features of hinge joints and their movements as discussed in class.
- Encourage students to think about examples of hinge joints in everyday life and how they impact movement.
- Mention how this foundational knowledge will lead into future lessons about different types of joints and their functions.
- End with a question-and-answer session to solidify understanding.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- KLB BK IV

**Learning Activities:**

- Recap discussion
- Question-and-answer session
- Encourage students to think of practical examples

**Teacher Self-Evaluation:**



**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: SKELETAL MUSCLES.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to describe features of skeletal muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson with a warm greeting to the students, establishing a positive learning environment. Introduce the topic of skeletal muscles, explaining their importance in the human body, particularly in movement and posture. Ask students if they can name any skeletal muscles they are familiar with to engage them.

**Resources:**

- Chalk board/whiteboard
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Brief discussion on familiar skeletal muscles.

**Lesson Content (30 minutes):**

In this section, delve into the characteristics of skeletal muscles. Discuss the following key points:

**1. Structure of Skeletal Muscles:**

- Striated appearance
- Multi-nucleated cells
- Voluntary control

**2. Function of Skeletal Muscles:**

- Facilitate movement by attaching to bones
- Maintain posture
- Generate heat during contraction

Introduce illustrative diagrams of skeletal muscles, highlighting key structures such as muscle fibers, tendons, and attachments to bones. Engage students in examining these diagrams, guiding them to observe and identify features discussed.

Organize a brief group discussion where students can share their thoughts on the function of skeletal muscles in everyday activities, such as exercising, lifting objects, or playing sports.

**Resources:**

- Illustrative diagrams of skeletal muscles (printed handouts or digital presentation)
- KLB Book IV, Pages 129-130

**Learning Activities:**

- Examine diagrams of skeletal muscles
- Group discussion on the functions of skeletal muscles

**Conclusion (5 minutes):**

Summarize the key points covered during the lesson, emphasizing the structure and function of skeletal muscles. Remind students of their role in movement and the importance of keeping themselves healthy and fit for maintaining strong skeletal muscles. Prepare the students for the next lesson, hinting at further exploration of muscle types and how they work together.

**Resources:**

- Chalk board/whiteboard
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Summary discussion and Q&A session

**Teacher Self-Evaluation:**

**TEACHER'S NAME:**\_\_\_\_\_ **TSC NO:**\_\_\_\_\_

**SCHOOL/INSTITUTION:**\_\_\_\_\_

**FORM:**4

**TERM:**3

**SUBJECT:** BIOLOGY

**TOPIC:** SUPPORT & MOVEMENT IN PLANTS AND ANIMALS

**SUB TOPIC:** SKELETAL MUSCLES.

**WEEK:**\_\_\_\_\_ **LESSON NO:**\_\_\_\_\_

**DATE**\_\_\_\_\_ **TIME**\_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe features of skeletal muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Greet the students and engage them with a quick question about their prior knowledge of muscles and their functions in movement. Briefly introduce the topic of skeletal muscles, emphasizing their importance in the body.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, PP 129-30)

**Learning Activities:**

- Whole-class discussion
- Question and answer session

**Lesson Content (30 minutes):**

Begin by explaining the characteristics and features of skeletal muscles including:

- Structure (striations, multi-nucleated)
- Function (voluntary movement, attachment to bones)
- Types of skeletal muscle fibers (slow-twitch vs. fast-twitch)

Display illustrative diagrams of skeletal muscles on the board or through digital means, guiding students in examining key features. Organize students into small groups to discuss the importance of these features in physical activities they engage in (e.g., sports, daily movements). Circulate to facilitate discussions and ensure understanding.



Conclude the content presentation by highlighting how the structural differences between the two types of muscle fibers influence their function in the body (e.g., endurance vs. strength activities).

**Resources:**

- Illustrative diagrams
- KLB BK IV, PP 129-30

**Learning Activities:**

- Group discussions
- Diagram examination
- Class sharing session

**Conclusion (5 minutes):**

Summarize the key points covered about the features of skeletal muscles, reiterating the significance of their structure and function in everyday movement and activities. Encourage students to think about how knowledge of skeletal muscles can help them in sports or other physical activities.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV, PP 129-30)

**Learning Activities:**

- Summary discussion
- Questions to check understanding

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: SKELETAL MUSCLES.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe features of skeletal muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson with a warm greeting to the students to create a positive learning environment.

Briefly introduce the topic of skeletal muscles, emphasizing their importance in the human body, especially their role in movement and support.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook (KLB BK IV, PP 129-30)
- Illustrative diagrams

**Learning Activities:**

- Overview of skeletal muscles
- Discussion on the importance and function of skeletal muscles

**Lesson Content (30 minutes):**

The teacher will present the detailed features of skeletal muscles, including their structure, functions, and unique characteristics.

*Key points to cover include:*

- Definition of skeletal muscles
- Location and attachment of skeletal muscles to bones
- Features: striations, multi-nucleation, and voluntary control
- Functions of skeletal muscles: movement, posture, and heat production

The teacher will use illustrative diagrams to depict the anatomy and arrangement of skeletal muscles. Students will examine these diagrams, identifying various parts and discussing their functions in pairs.

A brief discussion will also be held on how skeletal muscles are different from cardiac and smooth muscles. Encourage students to ask questions and share knowledge.

Conclude the content section with a recap of the critical features discussed.

**Resources:**

- KLB BK IV, PP 129-30
- Illustrative diagrams

**Learning Activities:**

- Examination of diagrams of skeletal muscles
- Group discussions on differences between skeletal, cardiac, and smooth muscles

**Conclusion (5 minutes):**

Summarize the key points covered in the lesson, reinforcing the features and functions of skeletal muscles. Take a moment to emphasize their significance in everyday activities and physical health. Prepare students by providing a brief overview of what will be covered in the next lesson.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook

**Learning Activities:**

- Summary discussion
- Preview of the next topic

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: SMOOTH OR VISCERAL MUSCLES.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe features of smooth muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson with a warm greeting to the students to create a positive learning environment. Introduce the topic of smooth muscles and explain their importance in the human body, particularly in involuntary movements. Provide a brief overview of what will be covered in the lesson.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 129-30)

**Learning Activities:**

- Teacher introduction and explanation
- Presentation of learning objectives

**Lesson Content (30 minutes):**

*1. Explanation and Features of Smooth Muscle:*

- Discuss the structure of smooth muscles, including their non-striated appearance, spindle shape, and single central nucleus.
- Explain the difference between smooth muscles and other muscle types (skeletal and cardiac).
- Highlight the involuntary nature of smooth muscles and their role in processes such as digestion, blood circulation, and respiratory functions.

*2. Examine Diagrams of Smooth Muscles:*

- Distribute impressively detailed diagrams of smooth muscle tissue for students to examine.
- Allow students to work in pairs to identify and label key features of smooth muscles on the diagrams.

- Encourage a brief class discussion about what they observed in the diagrams and how it correlates to the functions discussed.

### *3. Group Discussion:*

- Facilitate a group discussion centered around the functioning of smooth muscles in different systems of the body, such as the digestive and circulatory systems.
- Ask guided questions to promote critical thinking (e.g., “How do smooth muscles ensure efficient digestion?”).

### **Resources:**

- Illustrative diagrams of smooth muscle
- KLB BK IV. PP 129-30

### **Learning Activities:**

- Explanation of smooth muscle features
- Diagram examination
- Group discussion

### **Conclusion (5 minutes):**

Summarize the key points discussed during the lesson, emphasizing the distinct characteristics and involuntary functions of smooth muscles. Reinforce the importance of smooth muscles in the overall functioning of the body and encourage students to reflect on the interrelationship between muscle types. Inform the students about what will be covered in the next lesson, building anticipation.

### **Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 129-30)

### **Learning Activities:**

- Summary discussion
- Preparation for the next lesson

### **Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: SMOOTH OR VISCERAL MUSCLES.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to describe features of smooth muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students and engage them with a brief introduction to the topic of smooth muscles, highlighting their significance in the human body.
- Mention that today's lesson will specifically focus on the features of smooth muscles and how they differ from other muscle types.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- KLB BK IV, pages 129-130

**Learning Activities:**

- Brief overview lecture
- Introduction of illustrative diagrams

**Lesson Content (30 minutes):**

- Present the key features of smooth muscles (e.g., spindle-shaped cells, involuntary control, non-striated appearance, etc.), emphasizing their specific functions in body systems, such as digestion and blood vessel regulation.
- Use illustrative diagrams to visually explain the structure and arrangement of smooth muscles. Make sure to compare and contrast smooth muscles with skeletal and cardiac muscles to highlight their unique properties.
- Divide students into small groups and have them examine the diagrams provided. Each group should discuss the characteristics and functions of smooth muscles and prepare to share one key point with the class.
- Conduct a brief class discussion where groups share their observations, ensuring to address misconceptions or gaps in understanding.

**Resources:**

- Illustrative diagrams of smooth muscles
- KLB BK IV, pages 129-130

**Learning Activities:**

- Group work
- Class discussions
- Examination of diagrams

**Conclusion (5 minutes):**

- Recap the features of smooth muscles shared during the lesson. Highlight key terms and concepts that were covered, such as involuntary control and unique structural characteristics.
- Prepare students for the next lesson, reminding them of the importance of understanding smooth muscles in the context of overall health and bodily function.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- KLB BK IV, pages 129-130

**Learning Activities:**

- Summarization of key points
- Brief Q&A to clear any remaining doubts

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: CARDIAC MUSCLES.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe the features of cardiac muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

A brief introduction to cardiac muscles will be given, including their importance in the human body. Greet the students and engage them by asking if they can name any specific features of the heart or its muscles.

**Resources:**

Chalk board/white board

A piece of chalk/marker pen

KLB BK IV (Pp 130-1)

Illustrative diagrams

**Learning Activities:**

- Question and Answer session
- Introduction to resources

**Lesson Content (30 minutes):**

Begin with an overview of muscle tissue types, emphasizing the role of cardiac muscles in the circulatory system.

*1. Examine Diagrams of Cardiac Muscles (15 minutes):*

- Show illustrative diagrams of cardiac muscles, highlighting features such as striations, intercalated discs, and involuntary control.
- Ask students to identify the key features in pairs using their textbooks and diagrams.

*2. Brief Discussion (10 minutes):*

- Facilitate a whole-class discussion on the unique characteristics of cardiac muscles compared to skeletal and smooth muscles. Consider asking:
- What makes cardiac muscles essential for survival?



- How do the features of cardiac muscles support their functions?

*3. Topic Review (5 minutes):*

- Reinforce the key points discussed by summarizing the unique features of cardiac muscles using the illustrated diagrams.

**Resources:**

KLB BK IV (Pp 130-1)

Illustrative diagrams

Chalk board/white board

**Learning Activities:**

- Group examination of diagrams
- Whole-class discussion
- Summary of key features

**Conclusion (5 minutes):**

Summarize the key points covered in the lesson, particularly the features of cardiac muscles and their significance in the body. Prepare students for the upcoming lesson on muscle types by emphasizing how understanding cardiac muscles will help in learning about muscle contractions next.

**Resources:**

Chalk board/white board

A piece of chalk/marker pen

KLB BK IV (Pp 130-1)

**Learning Activities:**

- Summary and questioning
- Preparation for the next topic

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: CARDIAC MUSCLES.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe features of smooth muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students and briefly introduce the topic of cardiac muscles, emphasizing the connection to smooth muscles.
- Explain that today's lesson will focus on features of smooth muscles, which are essential to understanding overall muscle biology.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 130-1)

**Learning Activities:**

- Brief discussion on previous knowledge of muscle types
- Introduction to smooth muscle features

**Lesson Content (30 minutes):**

- Begin with examining illustrative diagrams of smooth muscles. Show different types of smooth muscle tissues and their locations in the body (e.g., blood vessels, digestive tract).
- Facilitate a discussion where students can share observations from the diagrams. Encourage them to note the distinct characteristics of smooth muscle, including:
  - Involuntary contraction
  - Non-striated appearance
  - Spindle-shaped cells
  - Ability to function for long periods without fatigue
- Discuss the physiological roles of smooth muscle and how they differ from cardiac and skeletal muscles.

- Engage students in a group activity where they work in pairs to list applications of smooth muscle in bodily functions, such as digestion, blood flow regulation, and respiration.
- Summarize key points during discussions and clarify any misconceptions.

**Resources:**

- Illustrative diagrams of smooth muscles
- Textbook (KLB BK IV. PP 130-1)

**Learning Activities:**

- Examine diagrams of smooth muscles
- Brief group discussions
- Topic review

**Conclusion (5 minutes):**

- Summarize the lesson by highlighting the key features of smooth muscles discussed, linking them back to their significance in the muscular system.
- Prepare students for the next lesson by giving a brief preview of what will be covered, which may include an exploration of comparisons between different muscle types.

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook (KLB BK IV. PP 130-1)

**Learning Activities:**

- Recap of smooth muscle features
- Brief Q&A session to reinforce learning

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_  
**SCHOOL/INSTITUTION:** \_\_\_\_\_  
**FORM:4**  
**TERM:3**  
**SUBJECT: BIOLOGY**  
**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**  
**SUB TOPIC: CARDIAC MUSCLES.**  
**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_  
**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe features of smooth muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin with a friendly greeting to create a positive learning environment. Introduce the lesson topic, explaining the significance of understanding cardiac and smooth muscles in the context of the human body and its movement. Briefly mention that today's focus is on smooth muscles.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook

**Learning Activities:**

- Brief discussion on prior knowledge of muscle types
- Introduction to the day's objectives

**Lesson Content (30 minutes):**

Engage students in examining illustrative diagrams of smooth muscles, emphasizing their structure and function. Discuss key features that distinguish smooth muscles from cardiac and skeletal muscles, including their involuntary nature, spindle-shaped cells, and location in organs.

*1. Examine Diagrams (10 minutes):*

- Provide students with illustrative diagrams of smooth muscle tissue.
- Have them identify and label different components, such as the nucleus and muscle fibers.

*2. Group Discussion (15 minutes):*

- Divide students into small groups to discuss their findings.
- Encourage them to share insights on the characteristics of smooth muscles and their role in various organ systems, such as the digestive and circulatory systems.

*3. Topic Review (5 minutes):*

- Come back together as a class and summarize the main points discussed.
- Ask guiding questions to facilitate a recap of what was learned regarding smooth muscles.

**Resources:**

- Illustrative diagrams of smooth muscles
- KLB BK IV, PP 130-1

**Learning Activities:**

- Diagram examination
- Group discussion and interaction
- Whole-class topic review

**Conclusion (5 minutes):**

Wrap up the lesson by highlighting the importance of smooth muscles and their unique characteristics compared to other muscle types. Reiterate the learning objectives and how they were met during the lesson. Briefly introduce what the next lesson will entail regarding cardiac muscles.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook

**Learning Activities:**

- Summary of lesson points
- Discussion of the next topic

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: CARDIAC MUSCLES.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe features of smooth muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

- Greet the students and briefly explain the relevance of understanding cardiac muscles in the broader context of biological systems.
- Introduce the specific focus on smooth muscles as part of muscular tissues.
- Help students connect prior knowledge with current learning by asking a few questions about muscle types.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook

**Learning Activities:**

- Brief discussion on previous knowledge of muscles.
- Introduction to the topic and objectives of the lesson.

**Lesson Content (30 minutes):**

- Initiate a detailed discussion on the features of smooth muscles including structure, function, and location in the body.
- Use illustrative diagrams to showcase the differences between smooth muscles and other muscle types.
- Facilitate a small group examination where each group analyzes specific diagrams of smooth muscle tissues.
- After group work, hold a brief discussion where each group shares its observations and findings.
- Encourage students to ask questions and clarify concepts about the muscle structure and its physiological role.

**Resources:**

- Illustrative diagrams
- KLB BK IV. PP 130-1
- Chalk board/white board

**Learning Activities:**

- Examine diagrams of smooth muscles in small groups.
- Group presentations to discuss findings.
- Whole-class discussion to reinforce understanding.

**Conclusion (5 minutes):**

- Recap the key features of smooth muscles discussed during the lesson.
- Encourage students to think about how this knowledge will connect to future lessons on muscular systems.
- Highlight the importance of understanding muscle types for comprehending bodily movements and health.

**Resources:**

- Chalk board/white board
- A piece of chalk/ marker pen
- Textbook

**Learning Activities:**

- Topic review and summary.
- Discussion on linkages to future topics.

**Teacher Self-Evaluation:**

**TEACHER'S NAME:** \_\_\_\_\_ **TSC NO:** \_\_\_\_\_

**SCHOOL/INSTITUTION:** \_\_\_\_\_

**FORM:4**

**TERM:3**

**SUBJECT: BIOLOGY**

**TOPIC: SUPPORT & MOVEMENT IN PLANTS AND ANIMALS**

**SUB TOPIC: CARDIAC MUSCLES.**

**WEEK:** \_\_\_\_\_ **LESSON NO:** \_\_\_\_\_

**DATE** \_\_\_\_\_ **TIME** \_\_\_\_\_

**Objectives:**

By the end of the lesson, the learner should be able to:

- Describe features of smooth muscles.

**LESSON PRESENTATION**

**Introduction (5 minutes):**

Begin the lesson with a warm greeting to create a positive learning environment. Introduce the topic of cardiac muscles and explain its relevance within the broader context of support and movement in living organisms. Engage students by asking questions about their prior knowledge of muscle types (e.g., "What do you know about different types of muscles?").

**Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

**Learning Activities:**

- Interactive questioning
- Overview of lesson objectives

**Lesson Content (30 minutes):**

*1. Discussion on Muscle Types (10 minutes)*

- Briefly review the three types of muscle tissue (skeletal, smooth, and cardiac) to set the context. Use diagrams to showcase the distinguishing features of each muscle type.

*2. Features of Cardiac Muscles (10 minutes)*

- Focus on the characteristics of cardiac muscles. Discuss the involuntary nature, the striated appearance, and the intercalated discs present in cardiac muscles. Use illustrative diagrams to explain how these features facilitate the function of the heart.

*3. Examine Diagrams (5 minutes)*



- Provide students with diagrams of smooth muscles. Allow them to examine these diagrams carefully and discuss the features that differentiate them from cardiac muscles.

#### *4. Brief Discussion (5 minutes)*

- Conduct a small group discussion where students share insights gained from examining the diagrams. Encourage them to summarize the characteristics of smooth and cardiac muscles, emphasizing their functions and importance in the human body.

Conclude this segment by asking students to reflect on their observations.

#### **Resources:**

- Illustrative diagrams of cardiac and smooth muscles
- KLB BK IV, PP 130-1

#### **Learning Activities:**

- Group discussions
- Examination of diagrams
- Q&A session

#### **Conclusion (5 minutes):**

Summarize the lesson by revisiting the main features of cardiac muscles discussed during the class. Highlight the importance of muscle types in the overall function of the human body and their implications in health and disease. Prepare students for the next lesson by asking them to think about how heart diseases may affect muscle function.

#### **Resources:**

- Chalk board/white board
- A piece of chalk/marker pen
- Textbook

#### **Learning Activities:**

- Summary discussion
- Questions for student reflection

#### **Teacher Self-Evaluation:**